ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 2 SET D **WINTER SEMESTER, 2024-2025**

DURATION: 25 MINUTES **FULL MARKS: 20**

MATH 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

1. Given, 2 + 2 (CO2) (PO1)

$$S = \{\begin{bmatrix} x^4 \\ y^{-2} \end{bmatrix} \mid \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2\}$$

This set represents all vectors that are scaled by a factor of 2 from the original vector space $\mathbb{R}^2$. **Is S a vector space?** If yes, justify your answer. If no, provide a counterexample to demonstrate why it is not.

Solution:

For S to be a vector space, it must satisfy the following properties:

1. For any two vectors $\mathbf{u}, \mathbf{v} \in S$, their linear combination $k_1\mathbf{u}+k_2\mathbf{v}$ must also be in S. (where k_1 and k_2 are scalars)
2. It must contain the zero vector.

Consider a vector for $(x, y) = (1, 1)$, which gives us the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Now, consider another vector for $(x, y) = (1, \frac{1}{2})$ which gives us the vector $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$, since $\left(\frac{1}{2}\right)^{-2} = 4$.

Now, if we take a linear combination of these two vectors, say

$$-2 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 3 \cdot \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} -5 \\ -14 \end{bmatrix},$$

we need to check if this resulting vector is in S.

But, there are no real values of x and y such that $x^4 = -5$ and $y^{-2} = -14$, since x^4 is always non-negative for any real x, and y^{-2} is always positive for any non-zero real y.

Therefore, the set S does not satisfy the closure property under linear combinations, and hence, it is **not a vector space**.

Alternative ways to show this:

- You can show this for any other vectors as well to prove the same point.
- You could have shown that scalar multiplication fails too.
- Showing that for $y = 0$ the vector becomes undefined.

Rubrics:

- 1 mark for pointing out that S is not a vectorspace.

- 1 mark for correct reasoning/explanation.
- 2 marks for providing a valid counterexample.

2. 16 (CO2) (PO1)

Given,

$$A = \begin{bmatrix} 1 & -1 & 1 & 0 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 2 & -2 & 2 & 1 & 5 \\ 1 & 0 & 3 & 1 & 3 \end{bmatrix} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

Answer the following questions (After reducing it to its Reduced Row Echelon Form (RREF)): (2×2)

(a) What is the dimension of Matrix A $(m \times n)$? (0.5)

(b) What is the rank of Matrix A? (0.5)

(c) Identify the pivot and free columns of Matrix A. (1)

(d) Find out the **solvability condition** for $Ax = b$. (1)

(e) For all the four fundamental subspaces associated with Matrix A (Columnspace $C(A)$, Nullspace $N(A)$, Rowspace $C(A^T)$, Left Nullspace $N(A^T)$), state the following: $(4 \times 2 + 2 \times 0.5)$

- Their dimensions.
- Bases of the subspaces.
- The subspaces they belong to (i.e., $\mathbb{R}^m$ or $\mathbb{R}^n$, or something else).
- Which two pairs are orthogonal/perpendicular to each other and justify by showing the dot product of their basis vectors. (showing one per pair is enough)

Solution:

We start with the augmented matrix $[A|b]$ based on the given system:

$$[A|b] = \left[\begin{array}{ccccc|c} 1 & -1 & 1 & 0 & 2 & b_1 \\ 0 & 1 & 2 & 1 & 1 & b_2 \\ 2 & -2 & 2 & 1 & 5 & b_3 \\ 1 & 0 & 3 & 1 & 3 & b_4 \end{array}\right]$$

Step 1: Eliminate entries below the first pivot in column 1. Perform $R_3 \to R_3 - 2R_1$ and $R_4 \to R_4 - R_1$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & 0 & 2 & b_1 \\ 0 & 1 & 2 & 1 & 1 & b_2 \\ 0 & 0 & 0 & 1 & 1 & b_3 - 2b_1 \\ 0 & 1 & 2 & 1 & 1 & b_4 - b_1 \end{array}\right]$$

Step 2: Eliminate entries below the second pivot in column 2. Perform $R_4 \to R_4 - R_2$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & 0 & 2 & b_1 \\ 0 & 1 & 2 & 1 & 1 & b_2 \\ 0 & 0 & 0 & 1 & 1 & b_3 - 2b_1 \\ 0 & 0 & 0 & 0 & 0 & b_4 - b_1 - b_2 \end{array}\right]$$

Step 3: Eliminate the entry above the third pivot in column 4. Perform $R_2 \to R_2 - R_3$:

$$\left[\begin{array}{ccccc|c} 1 & -1 & 1 & 0 & 2 & b_1 \\ 0 & 1 & 2 & 0 & 0 & b_2 - b_3 + 2b_1 \\ 0 & 0 & 0 & 1 & 1 & b_3 - 2b_1 \\ 0 & 0 & 0 & 0 & 0 & b_4 - b_1 - b_2 \end{array}\right]$$

Step 4: Eliminate the entry above the second pivot in column 2. Perform $R_1 \to R_1 + R_2$:

$$\text{RREF}([A|b]) = \left[\begin{array}{ccccc|c} 1 & 0 & 3 & 0 & 2 & 3b_1 + b_2 - b_3 \\ 0 & 1 & 2 & 0 & 0 & b_2 - b_3 + 2b_1 \\ 0 & 0 & 0 & 1 & 1 & b_3 - 2b_1 \\ 0 & 0 & 0 & 0 & 0 & b_4 - b_1 - b_2 \end{array}\right]$$

For the system to be consistent (solvable), the row of zeros in the coefficient matrix must correspond to a zero in the augmented column. This yields the condition (Answer to 2(d)):

$$\boxed{b_4 - b_1 - b_2 = 0 \quad \Rightarrow \quad b_4 = b_1 + b_2}$$

From the RREF form we can see that there are 3 pivot columns (1, 2, and 4) and 2 free columns (3 and 5). So, the rank of matrix A is 3 (Answer to 2(b)). For clarity, we can highlight the pivot columns in red and the free columns in green, and the pivots have been boxed (Answer to 2(c)):

$$\text{RREF}([A|b]) = \left[\begin{array}{ccccc|c} \boxed{1} & 0 & 3 & 0 & 2 & 3b_1 + b_2 - b_3 \\ 0 & \boxed{1} & 2 & 0 & 0 & b_2 - b_3 + 2b_1 \\ 0 & 0 & 0 & \boxed{1} & 1 & b_3 - 2b_1 \\ 0 & 0 & 0 & 0 & 0 & b_4 - b_1 - b_2 \end{array}\right]$$

Columnspace and Nullspace:

The bases of the columnspace of A are the original columns of A corresponding to the pivot columns (1, 2, and 4):

$$\text{Basis } C(A) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

The dimension of the columnspace is 3, and it is a subspace of $\mathbb{R}^4$.

As for the bases of its Nullspace, $N(A)$, we can write down $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$.

We set free variables as I and pivot variables as $-F$, and thus we get the bases of the Nullspace,

$$\text{Basis } N(A) = \left\{ \begin{bmatrix} -3 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Again, the transpose A^T is a 5×4 matrix:

$$A^T = \begin{bmatrix} 1 & 0 & 2 & 1 \\ -1 & 1 & -2 & 0 \\ 1 & 2 & 2 & 3 \\ 0 & 1 & 1 & 1 \\ 2 & 1 & 5 & 3 \end{bmatrix}$$

Step 1: Eliminate entries below the first pivot in column 1. Perform $R_3 \to R_3 - R_1, R_5 \to R_5 - 2R_1$:

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ -1 & 1 & -2 & 0 \\ 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Now perform $R_2 \to R_2 + R_1$ to get a pivot in row 2, column 2:

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Step 2: Eliminate entries below the second pivot in column 2. Perform $R_3 \to R_3 - 2R_2$, $R_4 \to R_4 - R_2$, $R_5 \to R_5 - R_2$:

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Step 3: Finalize RREF by swapping $R_3 \leftrightarrow R_4$ and clearing above the third pivot ($R_1 \to R_1 - 2R_3$, $R_5 \to R_5 - R_3$):

$$\text{RREF}(A^T) = \begin{bmatrix} \boxed{1} & 0 & 0 & 1 \\ 0 & \boxed{1} & 0 & 1 \\ 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

From the RREF form, we see 3 pivot columns (1, 2, and 3) and 1 free column (4). Thus, $\text{rank}(A^T) = 3$.

The basis for the rowspace of A (which is the columnspace of A^T) can be taken as three independent rows of A:

$$\text{Basis } C(A^T) = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 2 \\ 1 \\ 5 \end{bmatrix} \right\}$$

For the Left Nullspace $N(A^T)$, let $y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$. From the RREF of A^T we have

$$y_1 + y_4 = 0, \quad y_2 + y_4 = 0, \quad y_3 = 0,$$

so taking the free variable $y_4 = 1$ yields

$$\text{Basis } N(A^T) = \left\{ \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Answers:

(a)

The **Dimension** of A is 4×5.

(b)

The **Rank** of A is 3.

(c)

The **Pivot Columns** of A are: 1, 2, and 4. (highlighted in red)

The **Free Columns** of A are: 3, and 5. (highlighted in green)

$$\text{RREF}([A|b]) = \left[\begin{array}{ccccc|c} \boxed{1} & 0 & 3 & 0 & 2 & 3b_1 + b_2 - b_3 \\ 0 & \boxed{1} & 2 & 0 & 0 & b_2 - b_3 + 2b_1 \\ 0 & 0 & 0 & \boxed{1} & 1 & b_3 - 2b_1 \\ 0 & 0 & 0 & 0 & 0 & b_4 - b_1 - b_2 \end{array}\right]$$

(d)

The solvability condition for $Ax = b$ is:

$$\boxed{b_4 = b_1 + b_2}$$

(e)

Refer to the table below for the answers regarding the four fundamental subspaces associated with Matrix A.

(f)

The two pairs of orthogonal/perpendicular subspaces are:

- Columnspace $C(A) \perp$ Left Nullspace $N(A^T)$:

$$\begin{bmatrix} -1 \\ 1 \\ -2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix} = 1 - 1 + 0 + 0 = 0$$

  (The first vector is a basis vector of $C(A)$ and the second vector is the only basis vector of $N(A^T)$.)

- Nullspace $N(A) \perp$ Rowspace $C(A^T)$:

$$\begin{bmatrix} -3 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \\ 2 \end{bmatrix} = (-3) + 2 + 1 + 0 + 0 = 0$$

  (The first vector is a basis vector of $N(A)$ and the second vector is a basis vector of $C(A^T)$.)

Rubrics:

- 0.5 mark for correct dimension of A.
- 0.5 mark for correct rank of A.
- 1 mark for correctly identifying pivot and free columns.
- 1 mark for correct solvability condition.
- 8 marks (1 mark for bases, 0.5 for dimensions and 0.5 for subspaces for each subspace) for correctly stating the bases, dimensions, and subspaces for all four fundamental subspaces.
- 1 mark (0.5 mark each) for correctly identifying the orthogonal pairs and justifying with dot product calculations.
- 4 marks (2 marks each) for correct RREF calculations and steps.

	Columnspace $C(A)$	**Nullspace** $N(A)$
Bases	$\left\{\begin{bmatrix}1\\0\\2\\1\end{bmatrix}, \begin{bmatrix}-1\\1\\-2\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\\1\end{bmatrix}\right\}$	$\left\{\begin{bmatrix}-3\\-2\\1\\0\\0\end{bmatrix}, \begin{bmatrix}-2\\0\\0\\-1\\1\end{bmatrix}\right\}$
Dim	3	2
Subsp.	$\mathbb{R}^4$	$\mathbb{R}^5$

	Rowspace $C(A^T)$	**Left Nullspace** $N(A^T)$
Bases	$\left\{\begin{bmatrix}1\\-1\\1\\0\\2\end{bmatrix}, \begin{bmatrix}0\\1\\2\\1\\1\end{bmatrix}, \begin{bmatrix}2\\-2\\2\\1\\5\end{bmatrix}\right\}$	$\left\{\begin{bmatrix}-1\\-1\\0\\1\end{bmatrix}\right\}$
Dim	3	1
Subsp.	$\mathbb{R}^5$	$\mathbb{R}^4$